

Bernoulli Process

1.1 Definition

Definition 1.1. A **Bernoulli process** is an infinite sequence of independent and identically distributed (IID) Bernoulli random variables X_i :

$$\mathbb{P}(X_i = x) = p^x(1-p)^{1-x}, \text{ for } x \in \{0, 1\}.$$

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Discussion 1.1. Viewing each success in the Bernoulli process as an arrival, the Bernoulli process may be characterized in the following ways:

- i. The sequence of arrival times (T_k) , where T_k is the time of k th arrival.
- ii. The sequence of inter-arrival times (Δ_k) , where $\Delta_k = T_k - T_{k-1}$ is the time between the $(k-1)$ st and k th arrivals.
- iii. The number of arrivals by time k :

$$N_k = \max\{n \in \mathbb{N} : T_n \leq k\}.$$

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1.2 First Arrival Time

Proposition 1.1. The time to the first arrival in a Bernoulli process follows a geometric distribution:

$$\mathbb{P}(T_1 = n) = (1-p)^{n-1}p, \text{ for } n \in \mathbb{N}^{>0}$$

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Proof. The first arrival occurs at time n if and only if $X_1 = \dots = X_{n-1} = 0$ and $X_n = 1$. Since each arrival indicators (X_n) are IID:

$$\mathbb{P}(X_1 = \dots = X_{n-1} = 0, X_n = 1) = (1-p)^{n-1}p.$$

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Proposition 1.2. The geometric distribution is **memory-less**:

$$\mathbb{P}(T_1 > n+m | T_1 > m) = \mathbb{P}(T_1 > n).$$

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Proof. The unconditional probability that the time of first arrival exceeds n is:

$$\mathbb{P}(T_1 > n) = \mathbb{P}(X_1 = \cdots = X_n = 0) = (1 - p)^n.$$

The conditional probability that $T_1 > n + m$ given that $T_1 > m$ is:

$$\mathbb{P}(T_1 > n + m | T_1 > m) = \frac{\mathbb{P}(T_1 > n + m)}{\mathbb{P}(T_1 > m)} = \frac{(1 - p)^{n+m}}{(1 - p)^m} = (1 - p)^n.$$

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Proposition 1.3. The geometric distribution has constant hazard:

$$h(n) = \mathbb{P}(T_1 = n | T_1 \geq n) = p.$$

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Proof.

$$\mathbb{P}(T_1 = n | T_1 \geq n) = \frac{\mathbb{P}(T_1 = n)}{\mathbb{P}(T_1 \geq n)} = \frac{\mathbb{P}(T_1 = n)}{\mathbb{P}(T_1 > n - 1)} = \frac{(1 - p)^{n-1}p}{(1 - p)^{n-1}} = p.$$

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1.3 Arrival Times

Proposition 1.4. The number of arrivals by time n follows a binomial distribution:

$$\mathbb{P}(N_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \text{ for } k \in \{0, \dots, n\}.$$

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Proof. Let $(x_1, \dots, x_n)$ denote a length n binary sequence such that $\sum_{i=1}^n x_i = k$. Since the arrival indicators (X_i) are IID, the probability of any particular arrival sequence is:

$$\mathbb{P}\{(X_1, \dots, X_n) = (x_1, \dots, x_n)\} = p^k (1 - p)^{n-k}.$$

Now, there are $\binom{n}{k}$ distinct binary configurations $(x_1, \dots, x_n)$ that satisfy the constraint $\sum_{i=1}^n x_i = k$. Therefore:

$$\mathbb{P}(N_n = k) = \binom{n}{k} p^k (1 - p)^{n-k}.$$

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Proposition 1.5. The time of k th arrival in a Bernoulli process follows a negative binomial distribution:

$$\mathbb{P}(T_k = n) = \binom{n-1}{k-1} p^k (1-p)^{n-k}, \text{ for } n \geq k.$$

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Proof. The k th arrival occurs at time n if and only if the number of arrivals at time $n-1$ is $N_{n-1} = k-1$ and $X_n = 1$:

$$\mathbb{P}(T_k = n) = \mathbb{P}(N_{n-1} = k-1) \cdot \mathbb{P}(X_n = 1) = \binom{n-1}{k-1} p^{k-1} (1-p)^{n-k} \cdot p.$$

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1.4 Inter-arrival Times

Discussion 1.2. Define $\Delta_1 = T_1$ and $\Delta_k = T_k - T_{k-1}$. The Δ_k represent **inter-arrival times**. The time of k th arrival is, by construction, equal to the sum of the first k inter-arrival times:

$$\begin{aligned} T_k &= (T_k - T_{k-1}) + (T_{k-1} - T_{k-2}) + \cdots + (T_2 - T_1) + T_1 \\ &= \Delta_k + \Delta_{k-1} + \cdots + \Delta_2 + \Delta_1 = \sum_{i=1}^k \Delta_i. \end{aligned}$$

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Proposition 1.6. The inter-arrival times Δ_k are independent, and each follows a geometric distribution:

$$\mathbb{P}(\Delta_k = n) = (1-p)^{n-1} p, \text{ for } n \in \mathbb{N}^{>0}.$$

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Proof. In terms of the arrival indicators (X_i) , the k th inter-arrival time is:

$$\Delta_k = T_k - T_{k-1} = \#\{i : T_{k-1} < i \leq T_k\}.$$

Since these blocks of trials are non-overlapping and the arrival indicators are IID, the Δ_k are IID in turn. Now, $\Delta_k = n$ if and only if $X_{T_k} = 1$ and:

$$X_{T_{k-1}+1} = \cdots = X_{T_k-1} = 0$$

Thus,

$$\mathbb{P}(\Delta_k = n) = \mathbb{P}(X_{T_{k-1}+1} = \cdots = X_{T_k-1} = 0, X_{T_k} = 1) = (1-p)^{n-1} p.$$

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Proposition 1.7. The sequence of arrival times (T_k) has **stationary** and **independent** increments:

- i. For $k_0 < k_1$, $T_{k_1} - T_{k_0}$ has the same distribution as $T_{k_1 - k_0}$.
- ii. For $k_0 < k_1 < \dots$, the sequence of random variables:

$$(T_{k_0}, T_{k_1} - T_{k_0}, T_{k_2} - T_{k_1}, \dots)$$

is independent.

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1.5 Multinomial Process

Definition 1.2. Suppose that each arrival in a Bernoulli process may be classified into one of J mutually exclusive types, $X_i \in \{1, \dots, J\}$. Let $p_j = \mathbb{P}(X_i = j)$ denote the probability that an arrival is of type j . The parameters $(p_1, \dots, p_J)$ full characterize the distribution of X_i , and satisfy the constraints:

$$p_j > 0 \text{ for } j \in \{1, \dots, J\}, \text{ and } \sum_{j=1}^J p_j = 1.$$

The sequence of (X_j) is a **multinomial process**. Note that a non-arrival, coded as 0 in the Bernoulli process, may be regarded as one of the J arrival types. ■

Proposition 1.8. Let $N_n^{(j)}$ denote the number of type j arrivals by time n :

$$N_n^{(j)} = \sum_{i=1}^n \mathbb{I}(X_i = j).$$

The joint distribution of $(N_n^{(1)}, \dots, N_n^{(J)})$ is multinomial:

$$\mathbb{P}(N_n^{(1)} = n_1, \dots, N_n^{(J)} = n_J) = \binom{n}{n_1 \dots n_J} p_1^{n_1} \dots p_J^{n_J},$$

where $n = \sum_{j=1}^J n_j$ is the total number of arrivals. ◆

Proof. Let $(x_1, \dots, x_n)$ denote a sequence of length n such that:

$$\sum_{i=1}^n \mathbb{I}(X_i = j) = n_j, \text{ for } j \in \{1, \dots, J\}.$$

That is, the sequence $(x_1, \dots, x_J)$ contains n_1 elements equal to 1; n_2 elements equal to 2; and so forth. Since each arrival indicators (X_i) are IID, the probability of any particular arrival sequence is:

$$\mathbb{P}\{(X_1, \dots, X_n) = (x_1, \dots, x_n)\} = p_1^{n_1} \dots p_J^{n_J}.$$

Now, there are $\binom{n}{n_1 \dots n_J}$ distinct rearrangements of the sequence $(x_1, \dots, x_n)$ that leave the total number of arrivals of each type unchanged. Therefore:

$$\mathbb{P}(N_n^{(1)} = n_1, \dots, N_n^{(J)} = n_J) = \binom{n}{n_1 \dots n_J} p_1^{n_1} \dots p_J^{n_J}.$$

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Proof.

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Proposition 1.9. The marginal distribution of the number of arrivals by time n of type j is binomial:

$$\mathbb{P}(N_n^{(j)} = n_j) = \binom{n}{n_j} p_j^{n_j} (1 - p_j)^{n - n_j}, \text{ for } n_j \in \{0, \dots, n\}.$$

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1.6 Beta-Bernoulli Process

Definition 1.3. Suppose the probability of success P is itself random, following a beta distribution with parameters a and b :

$$f(p) = \frac{p^{a-1}(1-p)^{b-1}}{B(a, b)}, \text{ for } p \in (0, 1).$$

Given P , (X_i) is a sequence of conditionally independent indicator random variables:

$$(X_i \perp X_j) | P \text{ for } i \neq j,$$

each satisfying:

$$\mathbb{P}(X_i = 1 | P = p) = p.$$

Then, the sequence (X_i) is a **Beta-Bernoulli process**.

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Proposition 1.10. Let N_n denote the number of arrivals by time n , and let $x \in \{0, 1\}^n$ with $\sum_{i=1}^n x_i = k$. The *unconditional* probability of that sequence is:

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \frac{B\{a + k, b + (n - k)\}}{B(a, b)}.$$

Conditionally on $N_n = k$, every such sequence is equally likely:

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n | N_n = k) = \binom{n}{k}^{-1}.$$

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Proof. The unconditional probability follows by mixing the Bernoulli likelihood against the Beta prior:

$$\begin{aligned}\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) &= \mathbb{E}\{p^k(1-p)^{n-k}\} = \frac{1}{B(a, b)} \int_0^1 p^k(1-p)^{n-k} \cdot p^{a-1}(1-p)^{b-1} dp \\ &= \frac{B\{a+k, b+(n-k)\}}{B(a, b)}.\end{aligned}$$

There are $\binom{n}{k}$ sequences with k ones, each with this same probability, so the conditional distribution given $N_n = k$ is uniform on that set. ■

1.7 Exercises

- i. Prove that the sequence of arrival times in a Bernoulli process has stationary and independent increments.
- ii. For a Bernoulli process with $m \leq n$ and $j \leq k$, argue that:

$$\mathbb{P}(N_m = j | N_n = k) = \frac{\binom{m}{j} \binom{n-m}{k-j}}{\binom{n}{k}}, \text{ for } j \in \{0, \dots, k\}.$$

- iii. Prove that the marginal distribution of $N_n^{(j)}$ in a multinomial process is binomial.
- iv. Find the posterior expectation of P given $(X_1 = x_1, \dots, X_n = x_n)$ is a Beta-Bernoulli process.